

RESTAURANT APP AI APPROACH AND DATA FLOW

Prepared for: Restaurant Admin + Client App
Date: 5 March 2026

1) GOAL

Add AI features in a controlled and useful way for Admin + Client apps, without building a custom model on day one.

2) RECOMMENDED STRATEGY (BEST APPROACH)

Use: LLM API + RAG + Guardrails + Human Approval

- LLM API: text generation, summaries, recommendations
- RAG: answer only from your restaurant data
- Guardrails: scope control and safe outputs
- Human approval: admin confirms before final save/publish

3) CORE FEATURES TO LAUNCH

A. Admin Menu Content Assistant

- Generate item descriptions (short/long)
- Suggest tags (veg/non-veg, spicy, kids-friendly, bestseller)
- Translation support (Hindi/English)

B. Admin Image-Aware Suggestion

- After image upload, suggest dish name/category/tags
- Admin accepts or rejects suggestions

C. Restaurant-Only Admin Copilot

- Q&A for your own restaurant analytics and operations
- If data missing, AI must say "Not enough data"

D. Client Guided Search

- Natural queries like "spicy veg under 250"
- Recommend only currently available items

4) END-TO-END DATA FLOW

Flow A: Admin Description Generation

Admin UI -> Backend AI Endpoint -> Prompt Builder -> LLM -> Structured Output -> Backend Validation -> Admin Review -> Save DB

Flow B: Image Suggestion

Image Upload -> Metadata Prep -> AI Analysis -> Suggestions -> Admin Approval -> Save DB

Flow C: Restaurant-Only Copilot (RAG)

Admin Question -> Tenant Identification -> Context Retrieval -> Scoped Prompt -> LLM Response -> Source Trace -> UI Answer

5) DATA SOURCES FOR AI CONTEXT

- Menu items, categories, pricing, availability
- Order history and time-slot sales
- Restaurant timings, policies, charges, taxes
- Admin-maintained FAQ/policy content

6) SAFETY AND GOVERNANCE

- Strict tenant isolation (domain/id based)
- Scope-limited prompts (no out-of-business answers)
- PII masking before model call
- Rate limiting and request logging
- Fallback safe response on AI failure

7) COST AND PERFORMANCE

- Small model for simple writing tasks
- Strong model for analytics/copilot reasoning

- Cache repeated prompts
- Token budgets and output length limits

8) ROLLOUT PLAN

Phase 1 (fast):

- Admin description generator
- Image-based suggestions

Phase 2 (high value):

- Restaurant-only RAG copilot
- Daily analytics insight cards

Phase 3 (client intelligence):

- Natural language search
- Smart recommendation/upsell

9) SUCCESS METRICS

- Admin time saved per item creation
- AI acceptance rate
- Menu quality/completeness score
- Client conversion uplift
- Reduced repetitive support queries

10) FINAL RECOMMENDATION

Start with AI-assisted content + admin approval workflow,
then expand to restaurant-only RAG copilot.

This gives immediate value, lower risk, and scalable architecture.